Project:
Air Quality Data Analysis – Particulate Matter & Temperature Trends

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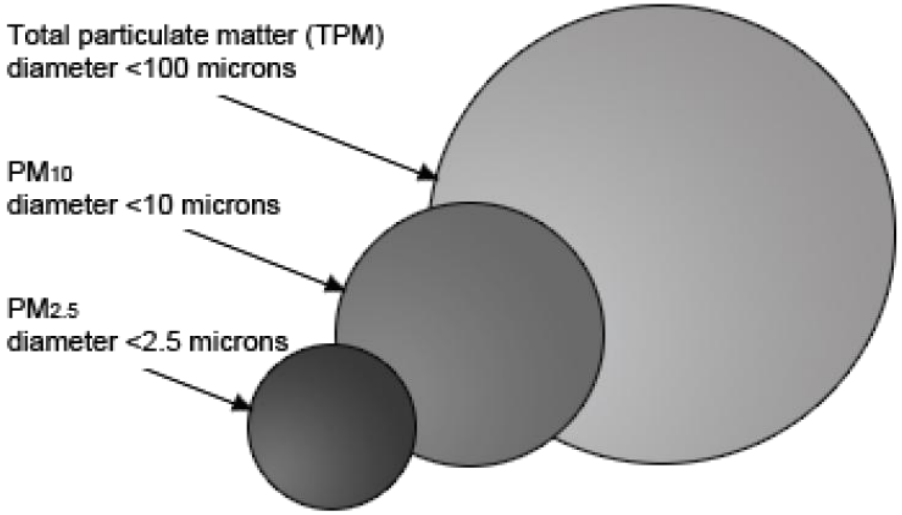
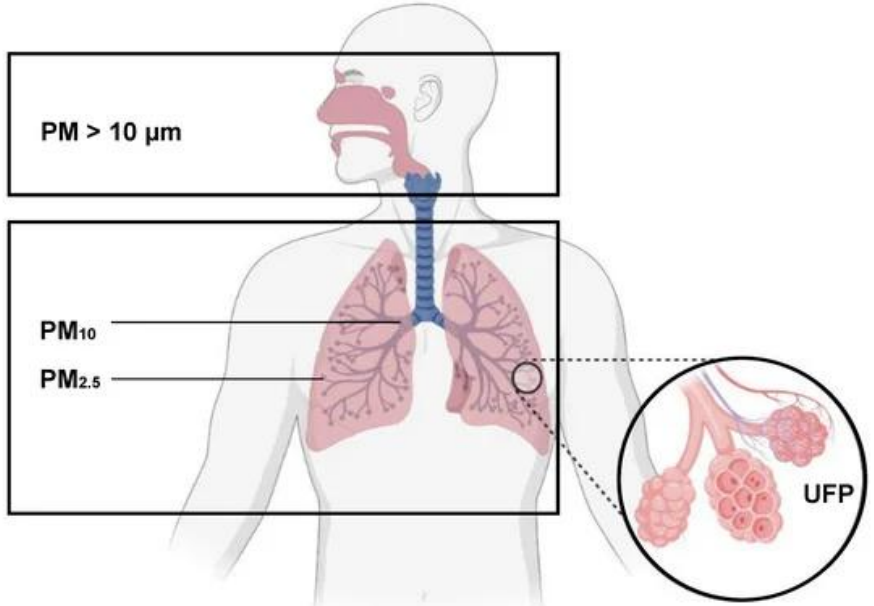
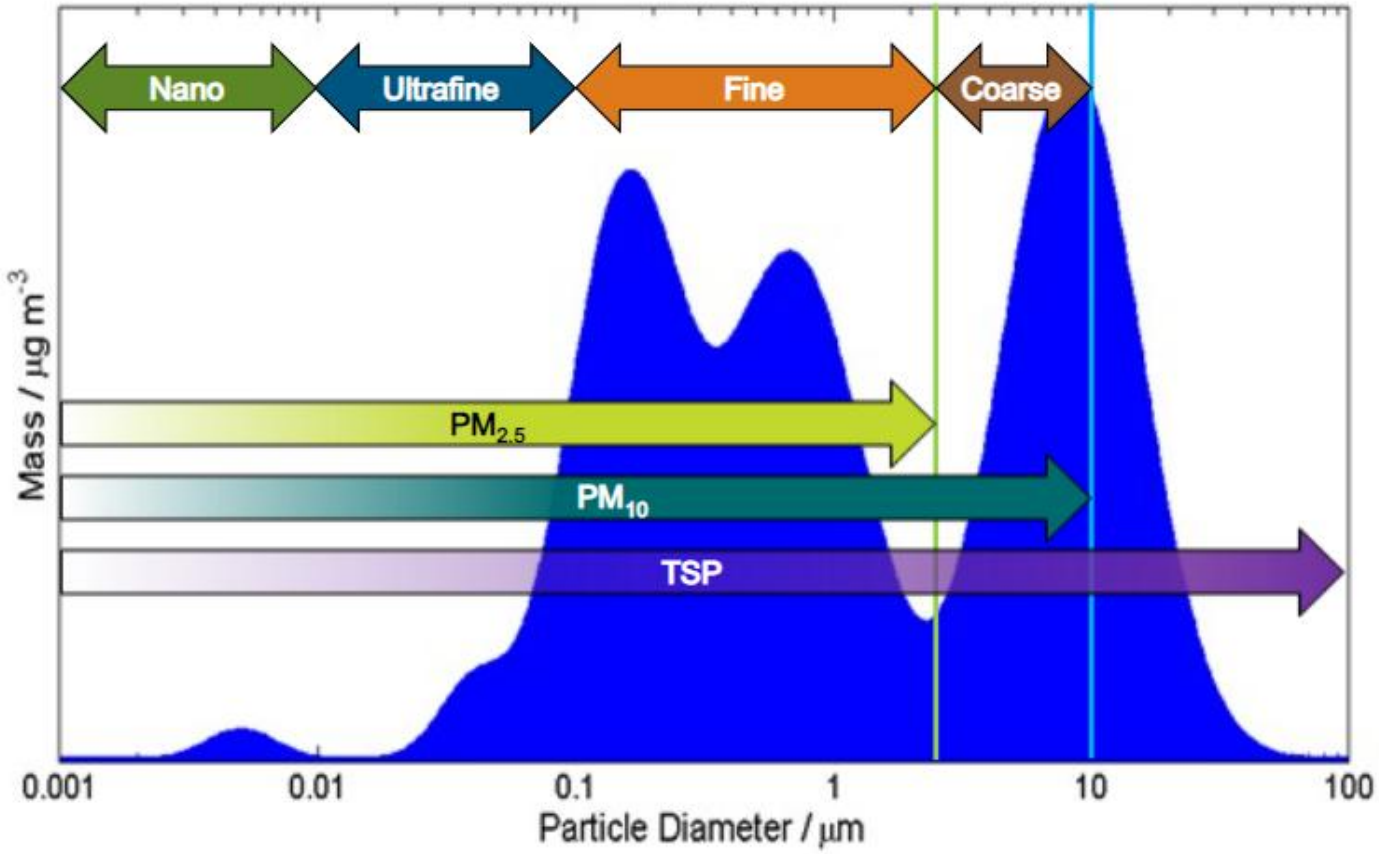
Introduction







Data Insights

Datasets: Input files

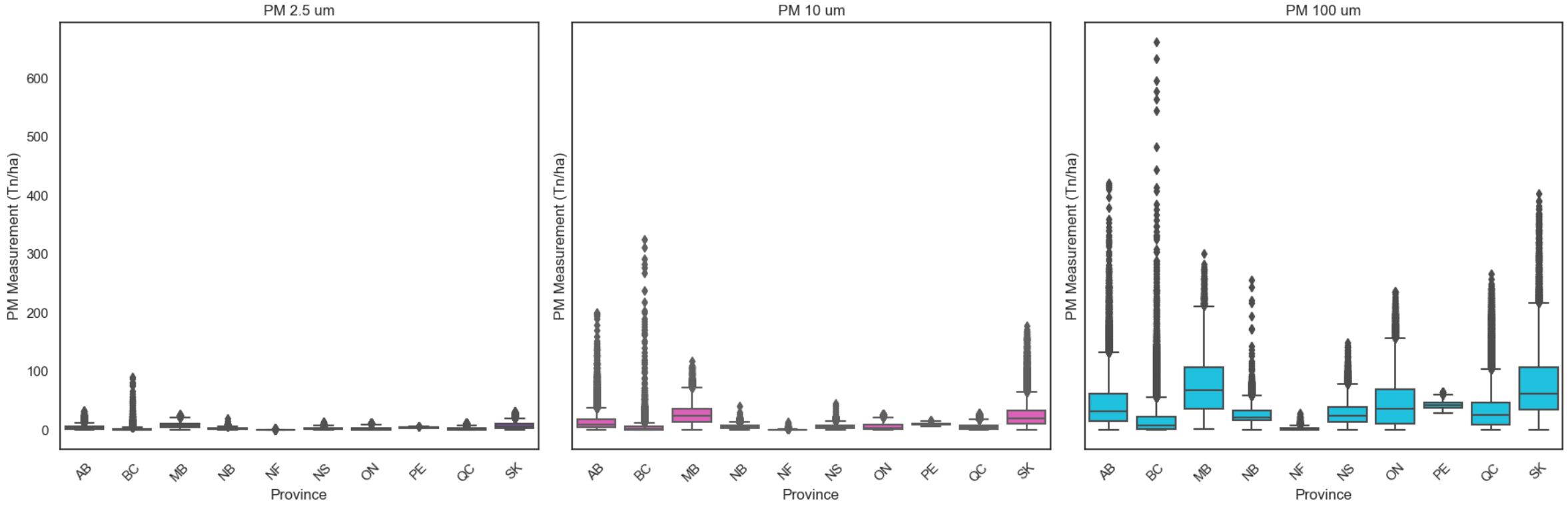
1. Air measurements data
2. Temperature data
3. Geolocation data x3

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 132331 entries, 0 to 132330
Data columns (total 17 columns):
 #   Column             Non-Null Count   Dtype
---  ------             --------------   -----
 0   SOIL_LANDSCAPE_ID  132331 non-null  int64
 1   PROVINCE           132331 non-null  object
 2   YEAR               132331 non-null  int64
 3   POLYGON_WEIGHT     132331 non-null  float64
 4   PROV_PCT           132331 non-null  float64
 5   PM10_VAL           132331 non-null  float64
 6   PM25_VAL           132331 non-null  float64
 7   TSP_VAL            132331 non-null  float64
 8   PM10_CLASS         132331 non-null  int64
 9   PM10_CLASS_EN      132331 non-null  object
 10  PM10_CLASS_FR      132331 non-null  object
 11  PM25_CLASS         132331 non-null  int64
 12  PM25_CLASS_EN      132331 non-null  object
 13  PM25_CLASS_FR      132331 non-null  object
 14  TSP_CLASS          132331 non-null  int64
 15  TSP_CLASS_EN       132331 non-null  object
 16  TSP_CLASS_FR       132331 non-null  object
dtypes: float64(5), int64(5), object(7)
memory usage: 17.2+ MB
None
(132331, 17)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 151 entries, 0 to 150
Data columns (total 12 columns):
 #   Column              Non-Null Count   Dtype
---  ------              --------------   -----
 0   time                151 non-null     object
 1   lat                 151 non-null     float64
 2   lon                 151 non-null     float64
 3   ssp126_tx_mean_p10  151 non-null     float64
 4   ssp126_tx_mean_p50  151 non-null     float64
 5   ssp126_tx_mean_p90  151 non-null     float64
 6   ssp245_tx_mean_p10  151 non-null     float64
 7   ssp245_tx_mean_p50  151 non-null     float64
 8   ssp245_tx_mean_p90  151 non-null     float64
 9   ssp585_tx_mean_p10  151 non-null     float64
 10  ssp585_tx_mean_p50  151 non-null     float64
 11  ssp585_tx_mean_p90  151 non-null     float64
dtypes: float64(11), object(1)
memory usage: 14.3+ KB
```

```
[3 rows x 41 columns]
<class 'geopandas.geodataframe.GeoDataFrame'>
RangeIndex: 3796 entries, 0 to 3795
Data columns (total 41 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   OBJECTID                3796 non-null   int64
 1   SOIL_LANDSCAPE_ID       3390 non-null   float64
 2   TSP_1981_VAL            3390 non-null   float64
 3   TSP_1981_CLASS          3390 non-null   float64
 4   TSP_1981_CLASS_EN       3390 non-null   object
 5   TSP_1981_CLASS_FR       3390 non-null   object
 6   TSP_1986_VAL            3390 non-null   float64
 7   TSP_1986_CLASS          3390 non-null   float64
 8   TSP_1986_CLASS_EN       3390 non-null   object
 9   TSP_1986_CLASS_FR       3390 non-null   object
 10  TSP_1991_VAL            3390 non-null   float64
 11  TSP_1991_CLASS          3390 non-null   float64
 12  TSP_1991_CLASS_EN       3390 non-null   object
 13  TSP_1991_CLASS_FR       3390 non-null   object
 14  TSP_1996_VAL            3390 non-null   float64
 15  TSP_1996_CLASS          3390 non-null   float64
 16  TSP_1996_CLASS_EN       3390 non-null   object
 17  TSP_1996_CLASS_FR       3390 non-null   object
 18  TSP_2001_VAL            3390 non-null   float64
 19  TSP_2001_CLASS          3390 non-null   float64
 20  TSP_2001_CLASS_EN       3390 non-null   object
 21  TSP_2001_CLASS_FR       3390 non-null   object
 22  TSP_2006_VAL            3390 non-null   float64
 23  TSP_2006_CLASS          3390 non-null   float64
 24  TSP_2006_CLASS_EN       3390 non-null   object
 25  TSP_2006_CLASS_FR       3390 non-null   object
 26  TSP_2011_VAL            3390 non-null   float64
 27  TSP_2011_CLASS          3390 non-null   float64
 28  TSP_2011_CLASS_EN       3390 non-null   object
 29  TSP_2011_CLASS_FR       3390 non-null   object
 30  TSP_2016_VAL            3389 non-null   float64
 31  TSP_2016_CLASS          3389 non-null   float64
 32  TSP_2016_CLASS_EN       3389 non-null   object
 33  TSP_2016_CLASS_FR       3389 non-null   object
 34  TSP_81_16_CHG_VAL       3389 non-null   float64
 35  TSP_81_16_CHG_CLASS     3389 non-null   float64
 36  TSP_81_16_CHG_CLASS_EN  3390 non-null   object
 37  TSP_81_16_CHG_CLASS_FR  3390 non-null   object
 38  Shape_Length            3796 non-null   float64
 39  Shape_Area              3796 non-null   float64
 40  geometry                3796 non-null   geometry
dtypes: float64(21), geometry(1), int64(1), object(18)
memory usage: 1.2+ MB
```

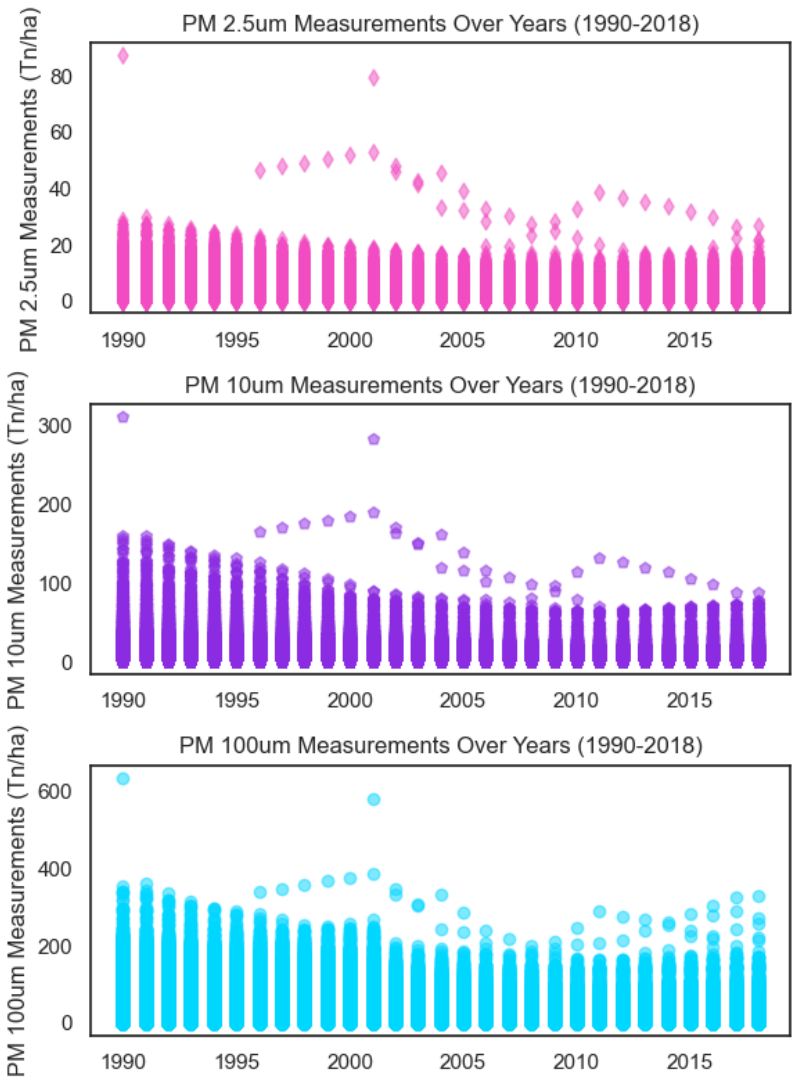
1- How is the different sizes PM emissions across Canada?



Higher air pollutants concentration in most of the provinces

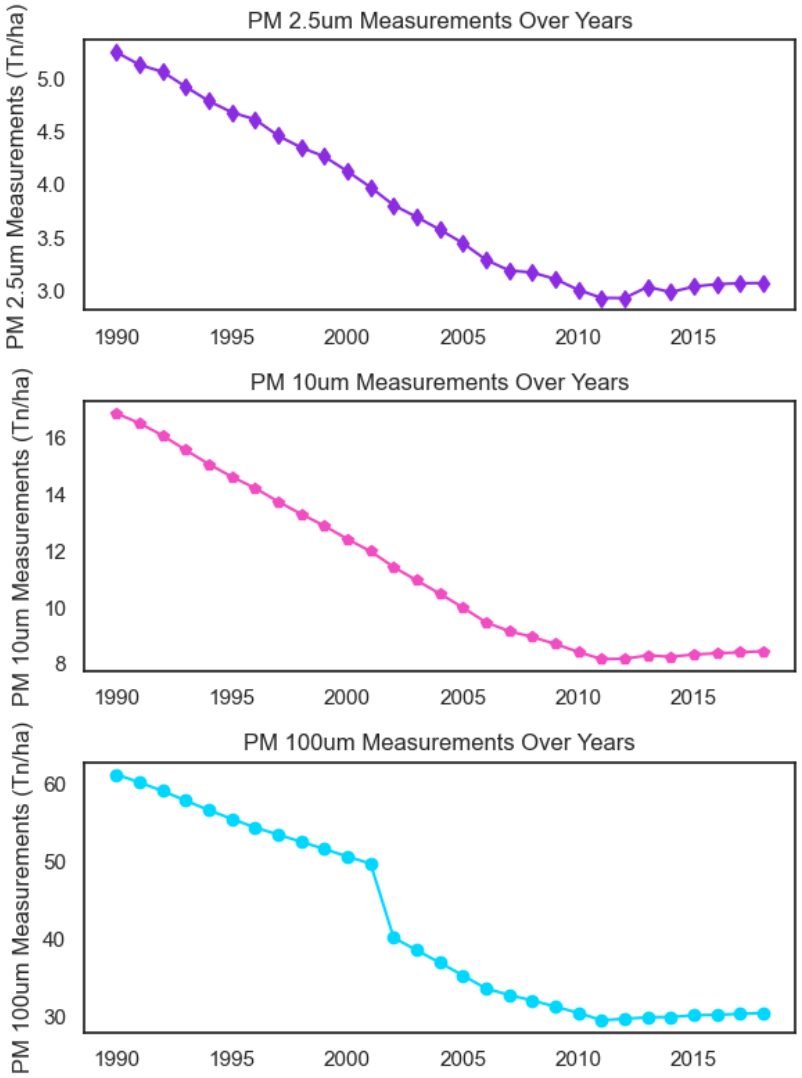
2- How were the PM emissions over the years?

Total



But looking at totals, there are values that remain high

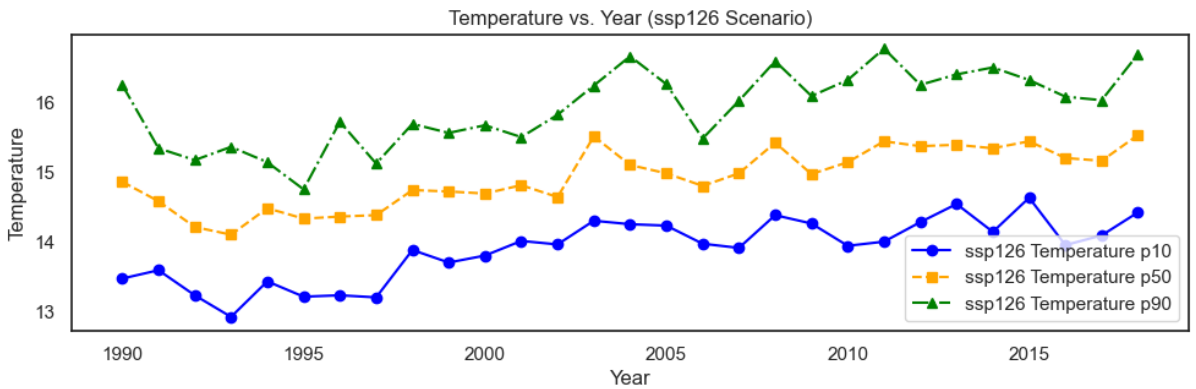
Average



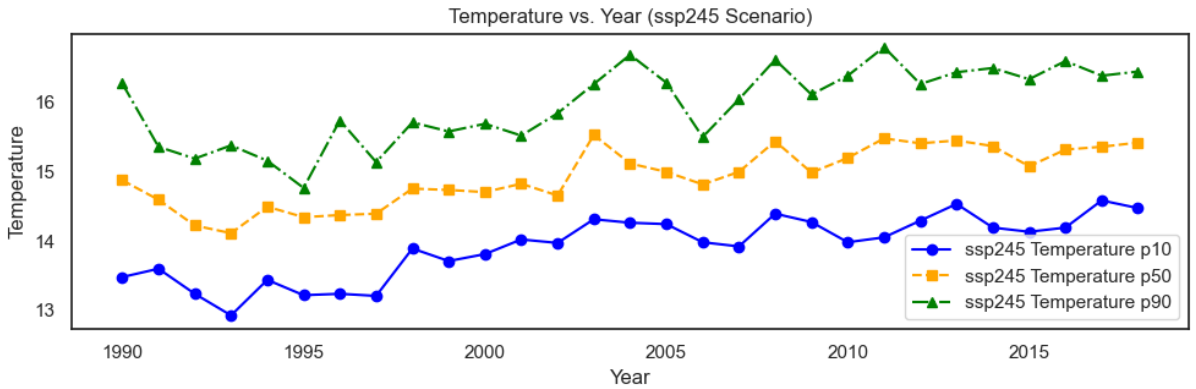
In average the PMs tend to decrease

3- How was the temperature variation over the years?

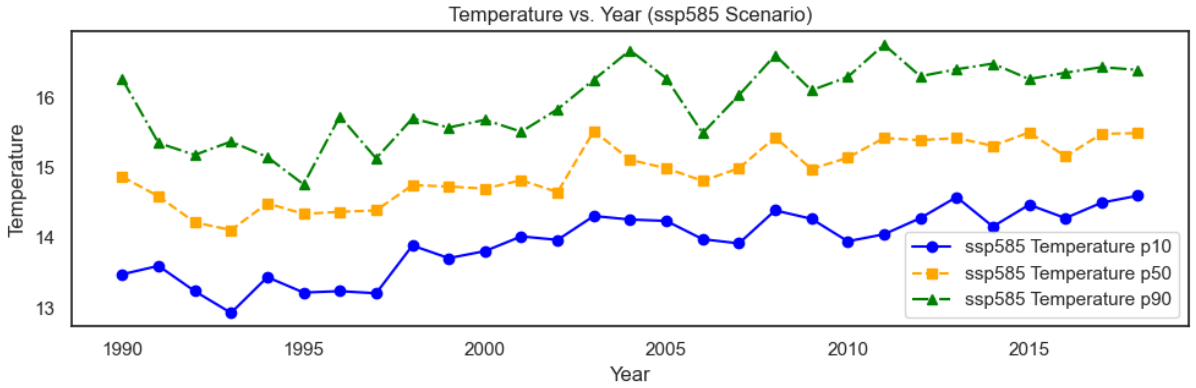
ssp126: Low environmental risks



ssp245: Intermediate environmental risks

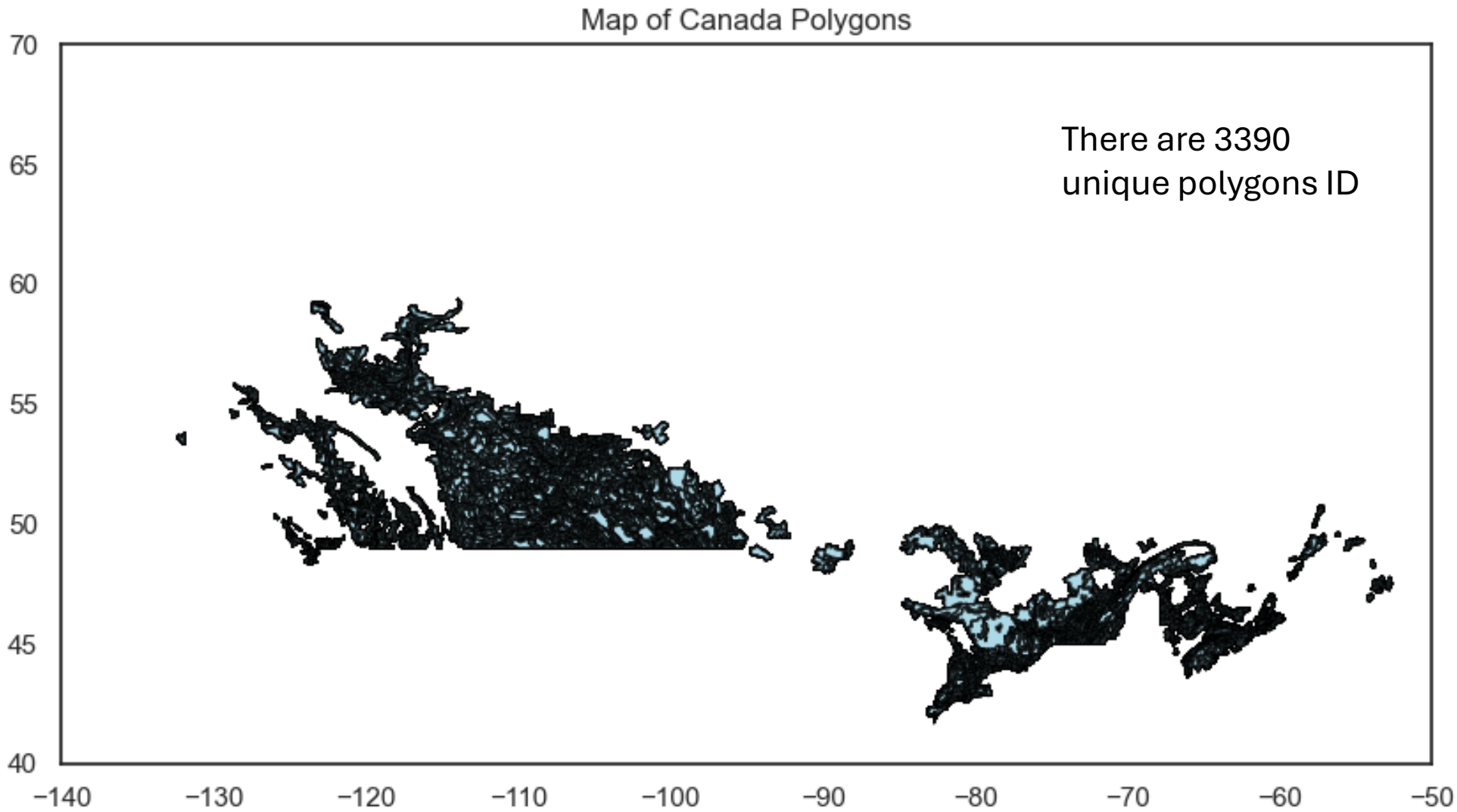


ssp585: High environmental risks

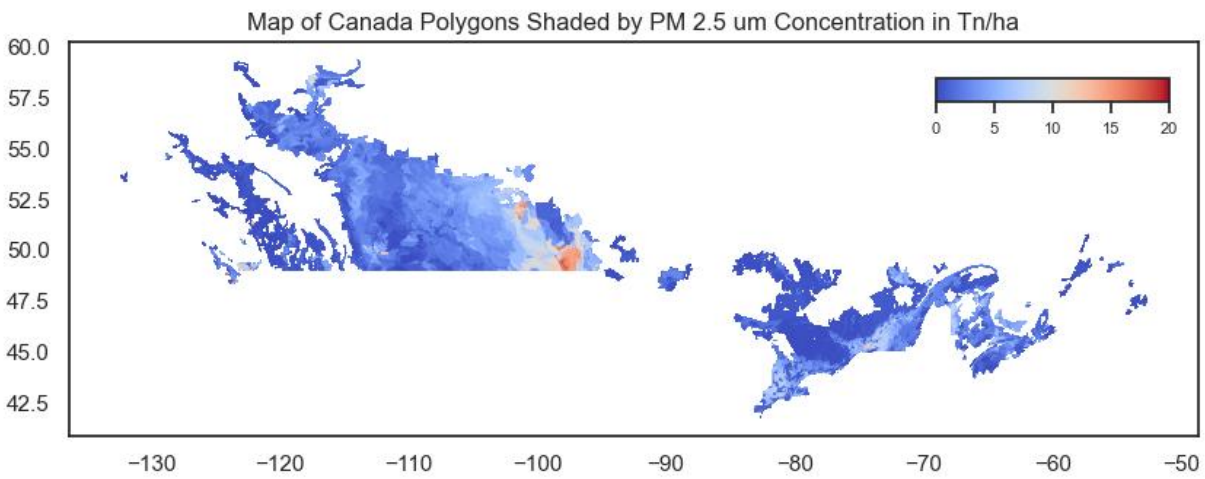


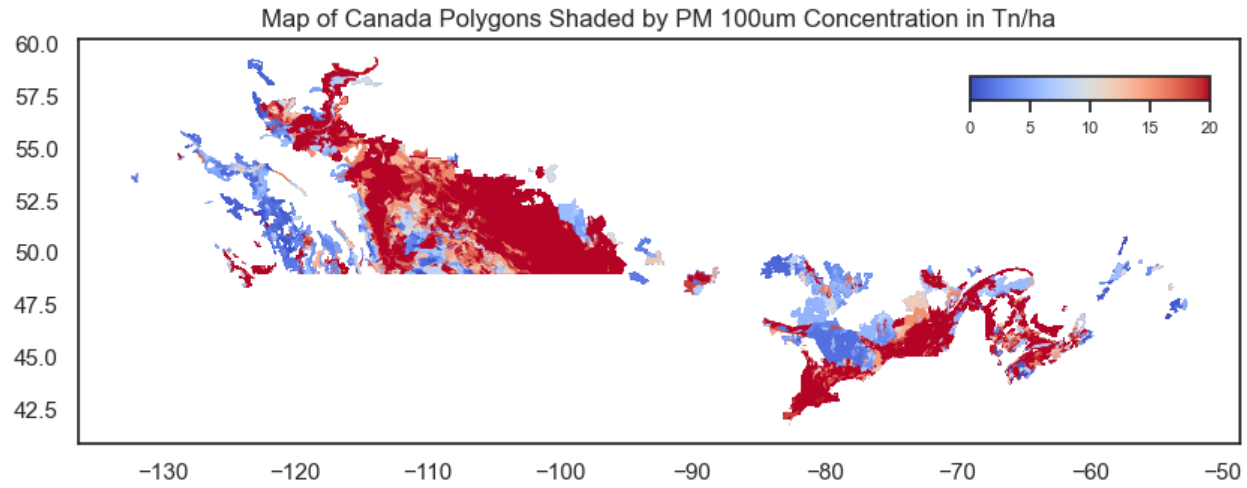
From 1990 to 2018, all temperature percentiles tend to increase

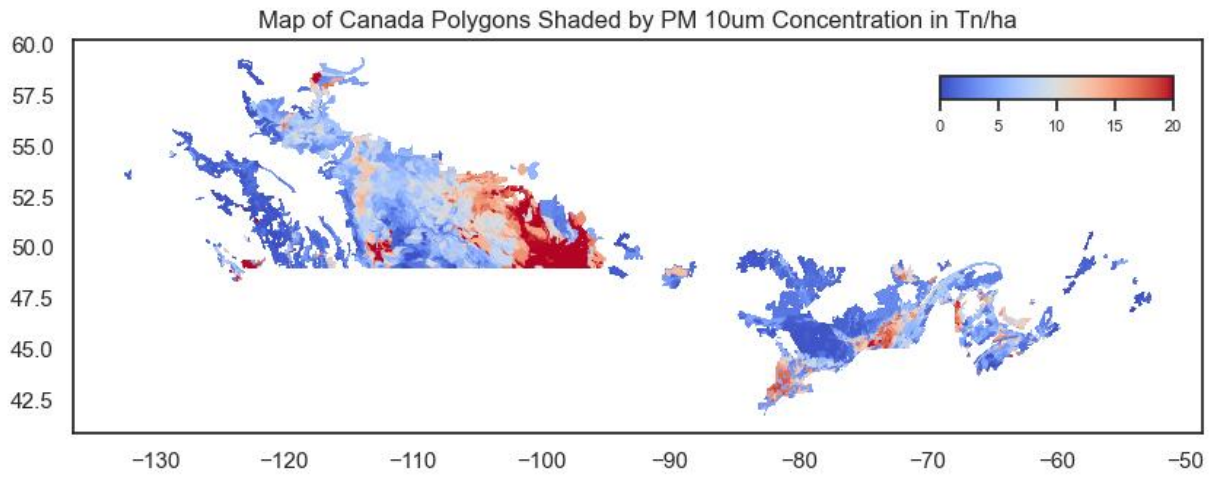
4- Where are those measures taken from?

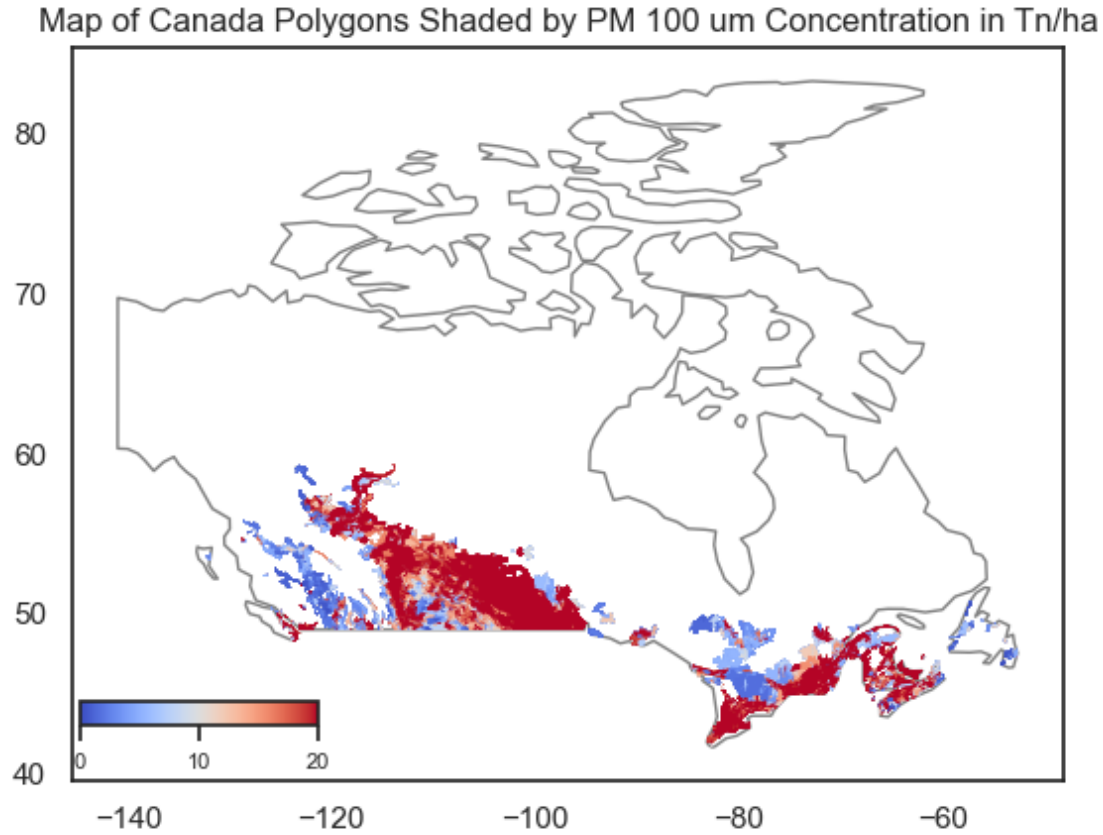


The higher PM measurements across Canada were for PM 100um









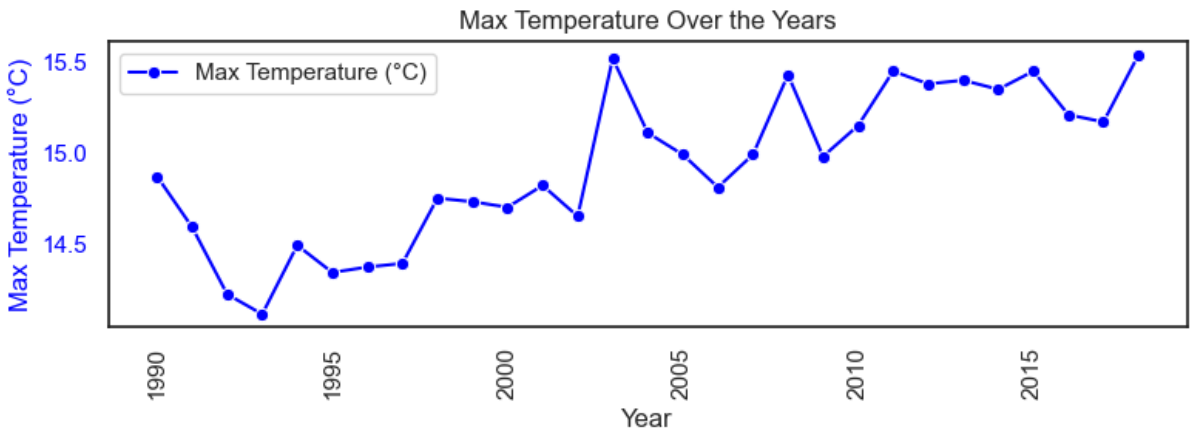
Datasets: Output file

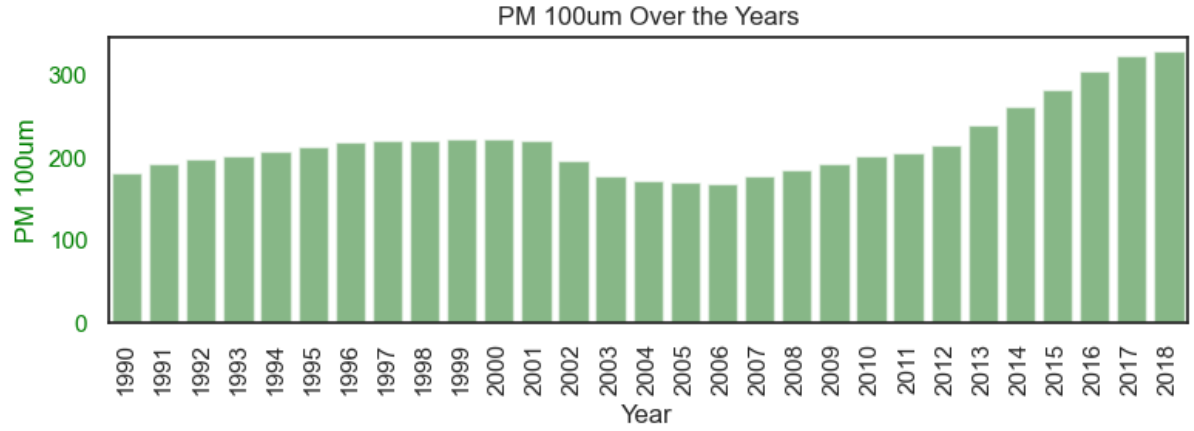
- Cleaning data
- Check for missing values
- Change data types
- Concatening the geofiles
- Check for unique polygon IDs
- Merging the differents tables
- Selection of those polygons ID that falls in the latitude and longitude of temperature table ID 959020
- Subsetting 1980-2018
- Merging the temperature table on year where ID 959020

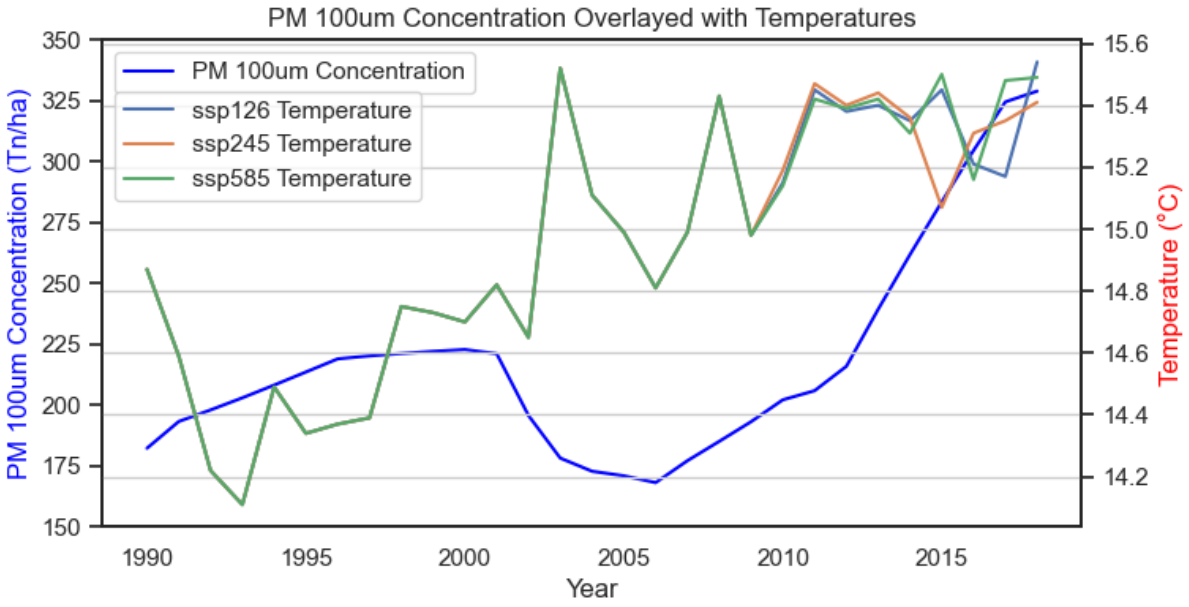
```
[29 rows x 28 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29 entries, 0 to 28
Data columns (total 28 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   ID                        29 non-null     int64
 1   Province                  29 non-null     object
 2   Year                      29 non-null     int64
 3   Polygon weight            29 non-null     float64
 4   Province %                29 non-null     float64
 5   PM 10um                   29 non-null     float64
 6   PM 2.5um                  29 non-null     float64
 7   PM 100um                  29 non-null     float64
 8   PM 10um classification    29 non-null     int64
 9   PM 10um class             29 non-null     object
 10  PM 2.5um classification   29 non-null     int64
 11  PM 2.5um class            29 non-null     object
 12  PM 100um classification   29 non-null     int64
 13  PM 100um class            29 non-null     object
 14  Shape length              29 non-null     float64
 15  Shape area                29 non-null     float64
 16  Geometry                  29 non-null     geometry
 17  lat                       29 non-null     float64
 18  lon                       29 non-null     float64
 19  ssp126_tx_mean_p10        29 non-null     float64
 20  ssp126_tx_mean_p50        29 non-null     float64
 21  ssp126_tx_mean_p90        29 non-null     float64
 22  ssp245_tx_mean_p10        29 non-null     float64
 23  ssp245_tx_mean_p50        29 non-null     float64
 24  ssp245_tx_mean_p90        29 non-null     float64
 25  ssp585_tx_mean_p10        29 non-null     float64
 26  ssp585_tx_mean_p50        29 non-null     float64
 27  ssp585_tx_mean_p90        29 non-null     float64
dtypes: float64(18), geometry(1), int64(5), object(4)
memory usage: 6.5+ KB
```

Output Insights

- How was the variation in temperature and TSP measurements?





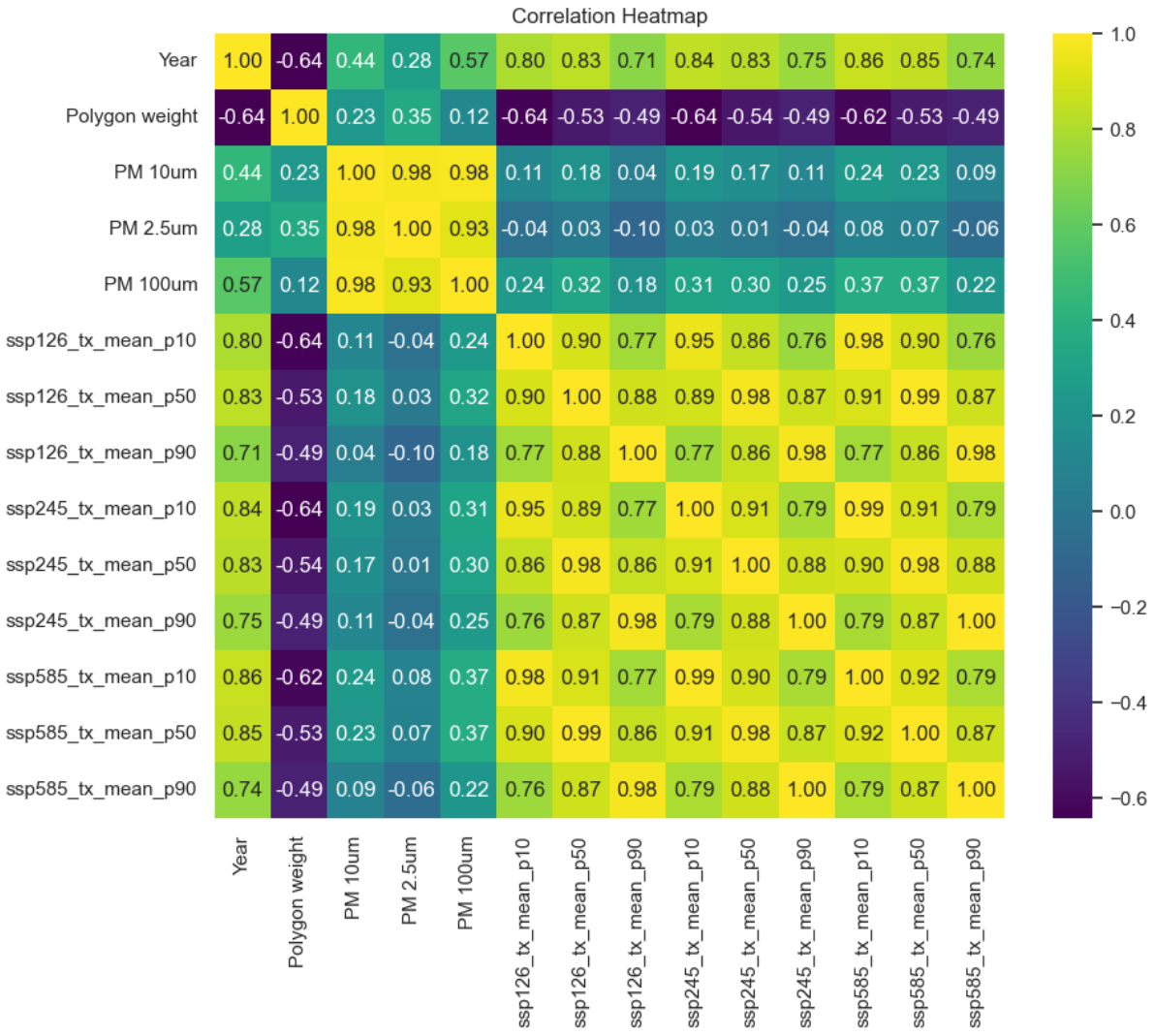


But the data also shows, in the recent years a tendency for high T and high PM 100um measures

The proven relation between PM 100um and T is:

- High T shows low PM 100um
- Low T shows high PM 100um

Analysis: Correlation



Study the relationship between the PM 100um air concentration and the 3 temperature scenarios ssp126, ssp245 and ssp585

```
Correlation between PM 100um and Year: 0.5743987897277943
Correlation between PM 100um and Polygon weight: 0.12440892400241702
Correlation between PM 100um and PM 10um: 0.9841199613878049
Correlation between PM 100um and PM 2.5um: 0.9291549081033871
Correlation between PM 100um and PM 100um: 1.0
Correlation between PM 100um and ssp126_tx_mean_p10: 0.2381060257309609
Correlation between PM 100um and ssp126_tx_mean_p50: 0.32240608127931947
Correlation between PM 100um and ssp126_tx_mean_p90: 0.1806228615633915
Correlation between PM 100um and ssp245_tx_mean_p10: 0.31411985710161866
Correlation between PM 100um and ssp245_tx_mean_p50: 0.30408061754857046
Correlation between PM 100um and ssp245_tx_mean_p90: 0.24566972323537006
Correlation between PM 100um and ssp585_tx_mean_p10: 0.36672538238022107
Correlation between PM 100um and ssp585_tx_mean_p50: 0.36552616154767714
Correlation between PM 100um and ssp585_tx_mean_p90: 0.2194925243507858
```

Analysis: Regression

sklearn.linear_model - LinearRegression

sklearn.ensemble - RandomForestRegressor

```
# Define independent variables (X=features) and dependent variable (y=target)
X = selected_numeric_columns[['Year', 'Polygon weight', 'PM 10um', 'PM 2.5um', 'ssp126_tx_mean_p10', 'ssp126_tx_mean_p50'
                    , 'ssp126_tx_mean_p90']]
y = selected_numeric_columns['PM 100um']
```

```
Mean Squared Error: 6.321779655040836
Coefficients: [ 5.57820878e-01  4.81892221e-03  1.38040900e+01 -1.84597363e+01
 -7.99910238e-01 -8.10852282e-01  1.03904891e+00]
Intercept: -1157.95255673279
R-squared: 0.9973101454096709
```

```
Mean Squared Error: 101.0475137827175
R-squared: 0.9570052843311214
```

```
# Define independent variables (X=features) and dependent variable (y=target)
X = selected_numeric_columns[['Year', 'Polygon weight', 'PM 10um', 'PM 2.5um', 'ssp245_tx_mean_p10', 'ssp245_tx_mean_p50'
                    , 'ssp245_tx_mean_p90']]
y = selected_numeric_columns['PM 100um']
```

```
Mean Squared Error: 5.835306652830069
Coefficients: [ 5.71517684e-01  4.99807086e-03  1.39001944e+01 -1.89284241e+01
 -6.42327668e-01  6.26627093e-01 -5.51636504e-01]
Intercept: -1182.3293359508484
R-squared: 0.9975171348508521
```

```
Mean Squared Error: 79.1774926088837
R-squared: 0.9663107615946517
```

```
# Define independent variables (X=features) and dependent variable (y=target)
X = selected_numeric_columns[['Year', 'Polygon weight', 'PM 10um', 'PM 2.5um', 'ssp585_tx_mean_p10', 'ssp585_tx_mean_p50'
                    , 'ssp585_tx_mean_p90']]
y = selected_numeric_columns['PM 100um']
```

```
Mean Squared Error: 6.009464776331253
Coefficients: [ 5.95177500e-01  5.14438983e-03  1.38787623e+01 -1.88813894e+01
 -7.96378785e-01 -6.61777798e-02 -1.15048244e-01]
Intercept: -1224.4908556561406
R-squared: 0.9974430322953212
```

```
Mean Squared Error: 82.18898459196711
R-squared: 0.9650294016016666
```

Predictions

With the values year:2035, Polygon weight:100, PM 10um:30 Tn/ha, PM 2.5um:8 Tn/ha, ssp245_tx_mean_p10: 13.5 C, ssp245_tx_mean_p50: 15 C and ssp245_tx_mean_p90: 15.5 C.

Which would be the value for PM 100um given those conditions?

```
Predicted PM 100um: [233.04274736]
```

```python
# Sample input data for prediction
sample_data = [[2025, 100, 30, 8, 18.5, 19, 19.5]]
# Make prediction
predicted_value = model.predict(sample_data)
print("Predicted PM 100um:", predicted_value)
```

```
Predicted PM 100um: [228.33594934]
```

```python
# Sample input data for prediction
sample_data = [[2025, 100, 30, 8, 10.5, 11, 11.5]]
# Make prediction
predicted_value = model.predict(sample_data)
print("Predicted PM 100um:", predicted_value)
```

```
Predicted PM 100um: [236.15678781]
```

The model predict values according with bibliography:
For high temperatures predict low particulate matter 100um air measures
For low temperatures predict high particulate matter 100um air measures

Conclusion

PM concentrations are influenced by various environmental factors, including temperature. Temperature can affect the dispersion, chemical reactions, and physical processes like evaporation and condensation of this particulate matter in the atmosphere.

Based on the analysis results and considering the relationship between PM 100um particulate matter concentrations in the air and the temperature variations, the following conclusion arise:

1- The predictive model developed demonstrates a strong capability to estimate PM 100um concentrations based on various environmental factors, including temperature.

2- The observed behavior of the model aligns with expectations regarding the impact of temperature on particulate matter concentrations.

3- When temperatures are higher, as indicated by the input features ssp245_tx_mean_p10, ssp245_tx_mean_p50, and ssp245_tx_mean_p90, the model predicts lower PM 100um concentrations. This outcome corresponds with existing scientific understanding, as higher temperatures can enhance atmospheric mixing and dispersion, leading to lower concentrations of particulate matter.

4- When temperatures are lower, the model predicts higher PM 100um concentrations. This relationship is consistent with the tendency for colder temperatures to decrease atmospheric mixing and dispersion, thereby allowing particulate matter to accumulate and persist in the air.

These findings underscore the intricate interplay between meteorological variables, such as temperature, and air quality parameters like particulate matter concentrations. Understanding these relationships is crucial for effective environmental management and public health interventions, especially in the context of climate change, where temperature variations may become more pronounced.

In summary, this project analysis highlights the significant impact of temperature variations on particulate matter concentrations in the air. By leveraging predictive modeling techniques, we can gain valuable insights into these relationships, aiding in the development of strategies to mitigate air pollution and safeguard public health in a changing climate.

Thank you